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Visual cues to reduce marine life mortality during a dredging operation

Abstract

A system for dredging a seabed includes a trailing suction hopper dredger. The trailing suction hopper dredger includes a hull, a rail coupled to the hull, and a dredge arm coupled to the rail. The dredge arm includes an upper suction pipe coupled to a lower suction pipe. A pump is coupled to the upper suction pipe and is configured to draw material toward the upper suction pipe from the seabed. A visual marine life deterrent is coupled to the dredge arm and is configured to direct light toward the sea floor where marine life may be located. Another visual marine life deterrent includes a silhouette of a marine life predator. The visual marine life deterrents are configured to be observable by the marine life and cause marine life to move away from the dredge arm.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6625907	12/2002	Murray	37/332	E02F 3/885
10167609	12/2018	Cashman	N/A	B63B 27/25
11557142	12/2022	Kerzner	N/A	G06V 20/41
2019/0069535	12/2018	Hauck, Jr.	N/A	A01M 29/10
2019/0150382	12/2018	Luzuriaga	N/A	E02B 15/041

OTHER PUBLICATIONS

Creel, P., "ERDC demonstrates new equipment, approach," U.S. Army, Aug. 18, 2016. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Patent Application No. 62/875,223, filed Jul. 17, 2019, the contents of which are incorporated herein in their entirety.

TECHNICAL FIELD

(1) The present invention relates generally to systems for reducing the mortality of sea turtles and other marine life during dredging operations.

BACKGROUND

(2) A trailing suction hopper dredger is a marine vessel used for dredging (e.g., removing) loose and soft soils such as sand, silt, gravel, or clay. One or more suction tubes equipped with a suction head are coupled to the vessel and are configured to reach the seabed. In a dredging operation using

a trailing suction hopper dredger, the suction head is lowered to the seabed and dragged along the seabed. The vessel's pump system draws material up from the seabed and loads the material into the hold of the trailing suction hopper dredger.

(3) During a dredging operation, the dredge arms and/or the suction mechanisms can contact marine life. For example, when contacted by the dredge arm and/or the suction mechanism, sea turtles can be harmed. There are a number of regulatory requirements and industry standards in place for marine operations in general for the purpose of reducing the amount of marine life harmed by the equipment being used. Similar requirements exist in the dredging industry to reduce harm caused to marine life during dredging operations.

SUMMARY

(4) In one set of embodiments, a system for dredging a seabed comprises a trailing suction hopper dredger. The trailing suction hopper dredger includes a hull in contact with a body of water, a rail coupled to the hull, and a dredge arm coupled to the rail. The dredge arm includes an upper suction pipe coupled to the rail and a lower suction pipe coupled to the upper suction pipe. A drag head is coupled to the lower suction pipe, the drag head in fluid communication with the lower suction pipe and configured to contact a seabed. A visual marine life deterrent is coupled to the dredge arm and is configured to direct light toward the sea floor where marine life may be located. The light is configured to be observable by the marine life, thereby influencing the marine life to move away from the dredge arm. A pump is coupled to the upper suction pipe and is in fluid communication with the upper suction pipe. The pump is configured to draw material toward the upper suction pipe from the seabed.

(5) In another set of embodiments a system for dredging a seabed comprises a rail configured to be coupled to a hull of a dredging vessel. A dredge arm is coupled to the rail and comprises an upper suction pipe coupled to the rail and a lower suction pipe coupled to the upper suction pipe. The dredge arm further comprises a drag head coupled to the lower suction pipe, the drag head in fluid communication with the lower suction pipe and configured to contact a seabed. A visual marine life deterrent is coupled to the dredge arm, the visual marine life deterrent comprising a silhouette of a marine life predator. The marine life predator is configured to be observable by the marine life, thereby influencing the marine life to move away from the dredge arm. A pump is coupled to the upper suction pipe. The pump is in fluid communication with the upper suction pipe and is configured to draw material toward the upper suction pipe from the seabed.

(6) In yet another set of embodiments, a system for dredging a seabed includes a trailing suction hopper dredger. The trailing suction hopper dredger includes a hull in contact with a body of water, and a dredge arm operatively coupled to the rail. The dredge arm includes an upper suction pipe operatively coupled to the rail and a lower suction pipe coupled to the upper suction pipe. A drag head is coupled to the lower suction pipe and is in fluid communication with the lower suction pipe and configured to contact the seabed. A marine life deterrent is coupled to the dredge arm and includes a physical marine life deterrent and a visual marine life deterrent. The physical marine life deterrent comprises a first end coupled to the dredge arm and a second end configured to hang freely from the first end. The visual marine life deterrent is coupled to the physical marine life deterrent and is configured to direct light based on an orientation of the physical marine life deterrent. A pump is coupled to the upper suction pipe and is in fluid communication with the upper suction pipe and is configured to draw material toward the upper suction pipe from the seabed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The details of one or more implementations are set forth in the accompanying drawings and the

description below. Other features, aspects, and advantages of the disclosure will become apparent from the description, the drawings, and the claims, in which:

(2) FIG. 1 is an illustration of a trailing suction hopper dredger.

(3) FIG. 2 is an illustration of a dredge arm for a trailing suction hopper dredger with a visual deterrent, according to a particular embodiment.

(4) FIG. 3 is an illustration of a dredge arm with another visual deterrent.

(5) FIG. 4 is an illustration of a dredge arm with yet another visual deterrent.

(6) FIG. 5 is an illustration of a dredge arm with a visual deterrent coupled to a physical deterrent.

(7) FIG. 6 is a detailed front view of the visual deterrent and the physical deterrent of the dredge arm of FIG. 5.

DETAILED DESCRIPTION

(8) Following below are more detailed descriptions of various concepts related to, and implementations of, a system for dredging a seabed. The various concepts introduced above and discussed in greater detail below may be implemented in any of numerous ways, as the described concepts are not limited to any particular manner of implementation. Examples of specific implementations and applications are provided primarily for illustrative purposes.

(9) I. Overview

(10) A trailing suction hopper dredger (“TSHD”) is configured to remove material from the sea floor via suction mechanisms attached to dredge arms. In some instances, the dredging process results in harm to marine life in the dredging location. Implementations herein relate to deterrents that function to reduce or eliminate harm to marine life resulting from the dredging process. In some embodiments, the deterrents include an array of lights, such as LED lights that are sufficiently powerful to be seen by marine life through the turbid waters caused by the dredging process. Other arrangements include one or more lights in communication with a reflector to amplify and distribute the effect of the lights. Silhouettes of predators can also be used to deter marine life, in some implementations.

(11) II. Conventional Trailing Suction Hopper Dredger

(12) FIG. 1 is an illustration of a TSHD **100**. The TSHD **100** is shown to include a dredge arm **102**, a rail **110**, and a hull **112**. The dredge arm **102** is rotatably and translationally coupled to the rail **110**, and the rail **110** is coupled to the hull **112**. In a deployed position, the dredge arm **102** extends below the hull **112**, as shown. To return to a stored position (not shown), the dredge arm **102** is hoisted toward the hull **112** until the dredge arm **102** is in a horizontal position. The dredge arm **102** is raised along the rail **110** until the dredge arm **102** reaches the deck of the TSHD **100** such that the dredge arm **102** can be secured until the next use.

(13) The dredge arm **102** further includes a lower suction pipe **104**, an upper suction pipe **106**, and a drag head **116**. The lower suction pipe **104** is coupled to the upper suction pipe **106** by an intermediate joint **118** (e.g., a gimbal joint) that provides for angular rotation of the lower suction pipe **104** relative to the upper suction pipe **106** in multiple planes. The components of the dredge arm **102** can be manufactured from any material suitable for use in marine dredging operations (e.g., metals, composites, or other materials suitable for the application). The upper suction pipe **106** is coupled to the rail **110** and provides a conduit through which material from the seabed **114** is transferred to the hull **112**. The pump **108** is located in the hull **112** and is coupled to the upper suction pipe **106**. The pump **108** is configured to direct material from the seabed **114** into the lower suction pipe **104** from the drag head **116**. The lower suction pipe **104** is coupled to the drag head **116** and provides a conduit through which material from the seabed **114** flows as it is directed toward the hull **112**. The drag head **116** is coupled to the lower suction pipe **104** and is configured to contact the seabed **114** and provide an aperture through which material from the seabed **114** is directed through the drag head **116** and into the lower suction pipe **104** as the material is directed toward the hull **112**.

(14) III. Example Dredge Arm with Light Array

(15) Referring to FIG. 2, an illustration of a dredge arm **202** with a visual deterrent is shown, according to a particular embodiment. The dredge arm **202** is substantially similar in construction to the dredge arm **102** of FIG. 1; however the dredge arm **202** includes additional components to act as a visual deterrent. The dredge arm **202** includes a first illumination device **204**, a second illumination device **206**, a third illumination device **208**, and a fourth illumination device **210** (collectively referred to herein as “illumination devices **204-210**”). It will be understood that any number of, and configuration of, illumination devices **204-210** can be used to achieve the purpose of deterring marine life, and the illumination devices **204-210** are shown as an exemplary embodiment. The first illumination device **204** emits a first illumination beam **212**, the second illumination device **206** emits a second illumination beam **214**, the third illumination device **208** emits a third illumination beam **216**, and the fourth illumination device **210** emits a fourth illumination beam **218** (collectively referred to herein as “illumination beams **212-218**”).

(16) The illumination devices **204-210** can be any type of illumination device that can be seen in turbid waters by marine life (e.g., a sea turtle **220**). In some embodiments, the illumination devices **204-210** are green LEDs emitting a light with a brightness of around 10,000 lumens. The illumination devices **204-210** can include LEDs or other appropriate lighting systems that emit light of any color within the visible range for the sea turtle **220** (e.g., about 300 nanometers to about 650 nanometers) and any other brightness that provides visible light through turbidity caused by the dredge arm **202**. The illumination devices **204-210** can direct the illumination beams **212-218** in a specific direction, in some implementations. For example, the illumination devices **204-210** can be floodlights that direct the illumination beams **212-218** away from the dredge arm **202** such that marine life moves away from the dredge arm **202**. The illumination devices **204-210** can also be arranged around the dredge arm **202** (e.g., in a 360 degree arrangement around the dredge arm **202**) to deter the sea turtle **220** from approaching the dredge arm **202** from all directions.

(17) IV. Example Dredge Arm with Light Reflector

(18) Referring to FIG. 3, an illustration of a dredge arm **302** with another visual deterrent is shown. The dredge arm **302** is substantially similar in construction to the dredge arm **102** of FIG. 1; however the dredge arm **302** includes additional components to act as a visual deterrent for the sea turtle **220**. The dredge arm **302** includes an illumination device **304** and a reflection device **308**. The illumination device **304** can be any type of illumination device with sufficient brightness to be seen in turbid waters by marine life (e.g., the sea turtle **220**) after being reflected. For example, the illumination device **304** can include a green LED emitting light with a brightness of approximately 10,000 lumens. The illumination device **304** can include LEDs or other appropriate lighting systems which emit light of any color within the visible range for the sea turtle **220** (e.g., about 300 nanometers to about 650 nanometers) and any other brightness that provides visible light after reflection through the turbidity caused by the dredge arm **302** during a dredging operation. The illumination device **304** includes an illumination beam **306** directed toward the reflection device **308**.

(19) The reflection device **308** is coupled to the dredge arm **302** and is configured to reflect light from the illumination beam **306** such that the light is directed away from the dredge arm **302**. In some embodiments, the reflection device **308** is coupled directly to the dredge arm **302**. In some arrangements, the reflection device **308** is coupled indirectly to the dredge arm **302**. For example, the reflection device **308** can be coupled to the dredge arm **302** via a flexible connector (e.g., a rope, cord, chain, etc.) or an inflexible connector (e.g., a rod, pipe, etc.) such that the reflection device **308** is spaced apart from the dredge arm **302**. In embodiments where the reflection device **308** is spaced apart from the dredge arm **302**, the orientation of the illumination device **304** and the illumination beam **306** are adjusted such that the light is directed toward the reflection device **308**.

(20) The reflection device **308** can include a reflective surface configured to reflect the light from the illumination beam **306**. In some arrangements, the reflection device **308** is a multi-faceted mirror positioned such that, when the light from the illumination beam **306** reaches the reflection

device **308**, the facets direct the light in a plurality of directions away from the dredge arm **302**, thereby creating a reflection beam **310** that can be seen by the sea turtle **220**. The reflection beam **310** can be the same brightness as the illumination beam **306** (e.g., approximately 10,000 lumens), in some implementations. The reflection device **308** can also be configured to amplify the brightness of the illumination beam **306**. In such configurations, the illumination beam **306** can be dimmer than approximately 10,000 lumens and the reflection device **308** can amplify the illumination beam **306** such that the reflection beam **310** is approximately 10,000 lumens.

(21) It will be understood that multiple illumination devices **304** and reflection devices **308** can be arranged around the dredge arm **302** (e.g., in a 360 degree pattern around the dredge arm **302**) such that multiple reflection beams **310** are directed away from the dredge arm **302** to deter the sea turtle **220** from approaching the dredge arm **302** from any direction during a dredging operation.

(22) V. Example Dredge Arm with Predator Proxy

(23) Referring to FIG. 4, an illustration of a dredge arm **402** with yet another visual deterrent is shown. The dredge arm **402** is substantially similar in construction to the dredge arm **102** of FIG. 1; however the dredge arm **402** includes additional components to act as a visual deterrent for the sea turtle **220**. The dredge arm **402** includes a shark silhouette **404** and a connector **406**.

(24) The shark silhouette **404** is configured to be of a similar size and shape as a living shark when viewed from the side. The shark silhouette **404** can be constructed of any material suitable to provide for realistic movement as the shark silhouette **404** moves through water (e.g., an undulating movement that mimics the movement of a shark in the water). An example of a suitable material for the shark silhouette **404** includes vinyl. The shark silhouette **404** is coupled to the dredge arm **402** via the connector **406**. The connector **406** can be a rigid connector (e.g., a rod, pipe, etc.), in some embodiments. The connector **406** can also be a flexible connector (e.g., a rope, cord, chain, etc.). In either instance, as the dredge arm **402** is pulled through the water by the TSHD **100**, the shark silhouette **404** is pulled through the water via the connector **406**. A shark is a natural predator of the sea turtle **220**, and seeing a shark in the water causes the sea turtle **220** to leave the area. Similarly, seeing the shark silhouette **404** near the dredge arm **402** causes the sea turtle **220** to leave the area around the dredge arm **402**. It will be understood that any number of shark silhouettes **404** can be arranged around the dredge arm **402** in various orientations to deter the sea turtle **220**.

(25) VII. Example Dredge Arm with Visual Deterrent and a Physical Deterrent

(26) Referring to FIG. 5, an illustration of a dredge arm **502** with a visual deterrent coupled to a physical deterrent is shown. The dredge arm **502** is substantially similar in construction to the dredge arm **102** of FIG. 1; however the dredge arm **502** includes additional components to act as a visual deterrent and a physical deterrent for the sea turtle **220**. The dredge arm **502** includes a first flexible component **504** and a second flexible component **506** (collectively referred to herein as “flexible components **504-506**”). In some embodiments, one or more of the flexible components **504-506** may include a plurality of flexible components arranged across a width of the draghead **116**. Such embodiments are described with reference to FIG. 6.

(27) Each of the flexible components **504-506** is coupled to the dredge arm **502** at a first end such that a second end is configured to hang freely from the first end. In some embodiments, the second end extends toward and contacts the seabed. The second end may also be configured such that a predetermined length of the second end remains in contact with the seabed when the dragarm **116** is in contact with the seabed. In example embodiments, the predetermined length is between two and ten feet, between three and nine feet, between four and eight feet, and between five and seven feet. As the TSHD **100** moves through the water, the second ends of the flexible components **504-506** move through the water. In some embodiments, one or more of the second ends contacts the sea turtle **220** as the TSHD **100** moves through the water. Upon contact from the second ends, the sea turtle **220** may move away from the dredge arm **502**, thereby preventing harm to the sea turtle **220**. Accordingly, the flexible components **504-506** may be constructed of any material suitable to be pulled through the water and contact the sea turtle **220**. The flexible components **504-506** may be

formed, for example, from metal (e.g., a chain constructed of at least two metal links where a first link is coupled to the dredge arm **502**, a second link is movably coupled to the first link, a third link is movably coupled to the second link etc., where each link hangs freely from the previous link in the chain). Regardless of the material from which the flexible components **504-506** are constructed, the flexible components **504-506** are configured to contact marine life (e.g., the sea turtle **220**) without crushing, pinning in place, exerting excessive force upon, or otherwise harming the marine life.

(28) Alternatively or additionally, the dredge arm **502** may act as a visual deterrent for the sea turtle **220**. In an example embodiment, the first flexible component **504** includes a first illumination device **508** and the second flexible component **506** includes a second illumination device **510**. The first illumination device **508** and the second illumination device **510** are collectively referred to herein as “illumination devices **508-510**.” The illumination devices **508-510** are substantially similar to the illumination devices **204-210** except that the illumination devices **508-510** are coupled to the flexible components **504-506** instead of the dredge arm **502**. Accordingly, the description of the various embodiments of the illumination devices **204-210** applies to the illumination devices **508-510**. The illumination devices **508-510** may be coupled directly to the flexible components **504-506** such that the first illumination device **508** is in physical contact with the first flexible component **504** and the second illumination device is in physical contact with the second flexible component **510**. The illumination devices **508-510** may also be coupled indirectly to the flexible components **504-506** such that the first illumination device **508** is not in physical contact with the first flexible component **504** and the second illumination device is not in physical contact with the second flexible component **510**. For example, the illumination devices **508-510** may be coupled to the flexible components **504-506** via a rope, a rod, or any other device or apparatus that is configured to couple the illumination devices **508-510** to the flexible components **504-506**.

(29) As the TSHD **100** moves through the water, the first illumination device **508** emits a first illumination beam **512** and the second illumination device **510** emits a second illumination beam **514** (collectively referred to herein as “illumination beams **512-514**”). As the flexible components **504-506** move through the water, the light from the illumination beams **512-514** is directed in various directions, where the various directions depend on the orientation of the flexible components **504-506**. When the sea turtle **220** sees the light from the illumination beams **512-514**, the sea turtle **220** moves away from the dredge arm **502**, thereby preventing harm to the sea turtle **220**.

(30) In some embodiments, the flexible components **504-506** can be included on any of the embodiments of the dredge arms **202**, **302**, **402**, and **502** to provide a physical deterrent in addition to the visual deterrents previously described. Though two flexible components are shown in the example embodiment, one of ordinary skill would understand that the dredge arm **502** can include more or fewer flexible components. Additionally, although each of the flexible components **504-506** is shown to include one illumination device, one or more of the flexible components **504-506** may include more than one illumination device.

(31) FIG. **6** is a detailed illustration of a front view of the visual deterrent and the physical deterrent of the dredge arm **502** of FIG. **5**. As shown, the flexible component **504** includes a first flexible element **606**, a second flexible element **608**, a third flexible element **610**, a fourth flexible element **612**, a fifth flexible element **614**, and a sixth flexible element **616** (collectively referred to herein as “flexible elements **606-616**”). Each of the flexible elements **606-616** may include, for example, a chain constructed of metal. Though six flexible elements are shown, one of skill in the art would appreciate that more or fewer flexible elements could be used. Each of the flexible elements **606-616** is coupled to an arm **604**, and the arm **604** is coupled to and extends from a retaining ring **602**. The arm **604** is rigid and is configured to support the flexible elements **606-616**. The retaining ring **602** is configured to rotate with respect to the dredge arm **502** when the arm **604** contacts an object.

For example, when raising the dredge arm **502** out of the water, the arm **604** may contact the hull **112**. In such instances, the retaining ring **602** rotates thereby preventing the arm **604** from damaging the hull **112**.

(32) The flexible elements **606-616** are spaced apart from each other by a distance d . In some embodiments, the distance d is approximately (e.g., within plus or minus ten percent) twelve inches. The flexible elements **606-616** are configured such that the distance D between the flexible element **606** and the flexible element **616** is greater than a width w of the draghead **116**. Arranged as described, the flexible elements **606-616** are configured to contact marine life as the draghead **116** moves along the seabed to move the marine life out of the way of the draghead **116**.

(33) Additionally and as shown, the illumination device **508** includes a first light **618** coupled to the first flexible element **606**, a second light **620** coupled to the second flexible element **608**, a third light **622** coupled to the third flexible element **610**, a fourth light **624** coupled to the fourth flexible element **612**, a fifth light **626** coupled to the fifth flexible element **614**, and a sixth light **628** coupled to the sixth flexible element **616**. The first light **618**, the second light **620**, the third light **622**, the fourth light **624**, the fifth light **626**, and the sixth light **628** are referred to herein as “lights **618-628**.” As the flexible elements **606-616** move through the water, the lights **618-628** move as well. Because each of the flexible elements **606-616** may encounter different water currents and/or turbulence and different objects located on the seabed, the flexible elements **606-616** may move independently of each other and be oriented differently than each other. Accordingly, the lights **618-628** may move independently of each other, thereby illuminating various portions of the water as the TSHD **100** moves through the water to motivate the marine life (e.g., the sea turtle **220**) to move out of the way of the dredge arm **502** and the drag head **116**.

(34) VI. Construction of Example Embodiments

(35) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed but rather as descriptions of features specific to particular implementations. Certain features described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(36) As utilized herein, the term “substantially,” “approximately,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

(37) The terms “coupled,” “attached,” and the like, as used herein, mean the joining of two components directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable or releasable). Such joining may be achieved with the two components or the two components and any additional intermediate components being integrally formed as a single unitary body with one another, with the two components, or with the two components and any additional intermediate components being attached to one another.

(38) It is important to note that the construction and arrangement of the system shown in the various example implementations is illustrative only and not restrictive in character. All changes and modifications that come within the spirit and/or scope of the described implementations are

desired to be protected. It should be understood that some features may not be necessary, and implementations lacking the various features may be contemplated as within the scope of the application, the scope being defined by the claims that follow. When the language a “portion” is used, the item can include a portion and/or the entire item unless specifically stated to the contrary. (39) Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

(40) Although only a few embodiments have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. For example, elements shown as integrally formed may be constructed of multiple components or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any method processes may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may also be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present invention.

Claims

1. A system for dredging a seabed, comprising: a trailing suction hopper dredger, the trailing suction hopper dredger comprising: a hull in contact with a body of water; a rail coupled to the hull; a dredge arm coupled to the rail, the dredge arm comprising: an upper suction pipe coupled to the rail; a lower suction pipe coupled to the upper suction pipe; a drag head coupled to the lower suction pipe, the drag head in fluid communication with the lower suction pipe and configured to contact the seabed; a visual marine life deterrent coupled to an intermediate location along the dredge arm via a connector, the connector including a flexible portion extending from the dredge arm, such that the visual marine life deterrent is movably suspended in the water between a sea floor and a surface of the water, the visual marine life deterrent configured to direct light toward the sea floor where marine life may be located, the light configured to be observable by the marine life, thereby influencing the marine life to move away from the dredge arm; and a pump coupled to the upper suction pipe, the pump in fluid communication with the upper suction pipe and configured to draw material toward the upper suction pipe from the seabed.
2. The system of claim 1, wherein the visual marine life deterrent comprises an illumination device.
3. The system of claim 2, wherein the light emitted by the illumination device has a brightness of approximately 10,000 lumens.
4. The system of claim 2, wherein the illumination device emits a green light.
5. The system of claim 2, wherein the illumination device comprises a light emitting diode.
6. The system of claim 2, wherein the illumination device emits a light that is of a wavelength that is visible to sea turtles.
7. The system of claim 2, further comprising a reflection device operatively coupled to the dredge arm, the reflection device operable to reflect a light beam from the illumination device.
8. A system for dredging a seabed, comprising: a rail configured to be coupled to a hull of a

dredging vessel; a dredge arm coupled to the rail, the dredge arm comprising; an upper suction pipe coupled to the rail; a lower suction pipe coupled to the upper suction pipe; a drag head coupled to the lower suction pipe, the drag head in fluid communication with the lower suction pipe and configured to contact a seabed; a visual marine life deterrent coupled to an intermediate location along the dredge arm, the visual marine life deterrent coupled to the intermediate location via a connector, the connector including a flexible portion extending from the dredge arm, such that the visual marine life deterrent is movably suspended within the water between a sea floor and a surface of the water, the visual marine life deterrent comprising a silhouette of a marine life predator, the marine life predator configured to be observable by marine life, thereby influencing the marine life to move away from the dredge arm; and a pump coupled to the upper suction pipe, the pump in fluid communication with the upper suction pipe and configured to draw material toward the upper suction pipe from the seabed.

9. The system of claim 8, wherein the marine life predator is a shark.

10. The system of claim 8, wherein the connector is a flexible connector coupling the visual marine life deterrent to the dredge arm.

11. The system of claim 8, wherein the connector includes a rigid connector coupling the visual marine life deterrent to the dredge arm.

12. A system for dredging a seabed, comprising: a trailing suction hopper dredger, the trailing suction hopper dredger comprising; a hull in contact with a body of water; a dredge arm operatively coupled to the hull, the dredge arm comprising; an upper suction pipe operatively coupled to the hull; a lower suction pipe coupled to the upper suction pipe; a drag head coupled to the lower suction pipe, the drag head in fluid communication with the lower suction pipe and configured to contact the seabed; a marine life deterrent coupled to the dredge arm, the marine life deterrent comprising: a physical marine life deterrent including a first end coupled to the dredge arm and a second end configured to hang freely from the first end; and a visual marine life deterrent coupled to the physical marine life deterrent and an intermediate location along the dredge arm via a connector, the connector including a flexible portion extending from the dredge arm, such that the visual marine life deterrent is movably suspended in the water between a sea floor and a surface of the water, the visual marine life deterrent configured to direct light based on an orientation of the physical marine life deterrent; and a pump coupled to the upper suction pipe, the pump in fluid communication with the upper suction pipe and configured to draw material toward the upper suction pipe from the seabed.

13. The system of claim 12, wherein the visual marine life deterrent is in contact with the physical marine life deterrent.

14. The system of claim 12, wherein the visual marine life deterrent comprises an illumination device.

15. The system of claim 12, further comprising a rail coupled to the hull, wherein the dredge arm is coupled to the rail.

16. The system of claim 14, wherein the illumination device emits a green light.

17. The system of claim 14, wherein the illumination device comprises a light emitting diode.

18. The system of claim 14, wherein the light emitted by the illumination device has a brightness of approximately 10,000 lumens.

19. The system of claim 18, wherein the illumination device emits a light that is of a wavelength that is visible to sea turtles.

20. A system for dredging a seabed, comprising: a trailing suction hopper dredger, the trailing suction hopper dredger comprising; a hull in contact with a body of water; a dredge arm operatively coupled to the hull, the dredge arm comprising; an upper suction pipe operatively coupled to the hull; a lower suction pipe coupled to the upper suction pipe; a drag head coupled to the lower suction pipe, the drag head in fluid communication with the lower suction pipe and configured to contact the seabed; a marine life deterrent coupled to the dredge arm, the marine life deterrent

comprising: a physical marine life deterrent including a first end coupled to the dredge arm and a second end configured to hang freely from the first end, wherein the physical marine life deterrent further comprises: a retaining ring rotatably coupled to the dredge arm; an arm coupled to and extending from the retaining ring; and a flexible element coupled to the arm; a visual marine life deterrent coupled to the physical marine life deterrent, the visual marine life deterrent configured to direct light based on an orientation of the physical marine life deterrent; and a pump coupled to the upper suction pipe, the pump in fluid communication with the upper suction pipe and configured to draw material toward the upper suction pipe from the seabed.
